

Phase 4 – Reporting and Insights

Model Comparison

K-Means: Silhouette Score = 0.7829, Davies-Bouldin Index = 0.2996, Calinski-Harabasz Score = 1293.44

Hierarchical Clustering: Silhouette Score = 0.7829, Davies-Bouldin Index = 0.2996

Interpretation: The clustering results indicate well-defined and separated clusters.

Neural Network Results

Confusion Matrix: [[18, 9], [6, 27]]

Accuracy: 75%, Precision: 75%, Recall: 81.82%, F1 Score: 78.26%

Conclusion

Clustering helps identify patterns, while Neural Networks are effective for prediction.

Insights

1. Clear delivery time segments exist.
2. Distance significantly affects delivery time.
3. Traffic conditions impact delivery delays.
4. Operational inefficiencies are observed.

Recommendations

1. Optimize routes using shortest path algorithms.
2. Assign delivery agents dynamically.
3. Manage peak-hour demand effectively.
4. Improve prediction models with more features.